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of Exeter

TM & NLP (COMM040)

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Intended Learning Outcomes



- Understand the strengths, weaknesses and biases of different NLP methods
- Clean, parse, transform and process large collections of text documents
- Know about recent advances and applications of NLP methods, their impact and real world applications.
- Communicate the pros and cons and trade-offs of different approaches to text analysis

Outline

- Introduction to IR
- IR pipeline
- RAG
- IR Evaluation

Outline

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- IR pipeline
- RAG
- IR Evaluation

Introduction to IR

- User has an **information need** and has some **collection of documents**
- User wants to find a relevant document

Introduction to IR

Google

Search bar with icons for search, voice search, image search, and AI Mode.

Google Search

I'm Feeling Lucky

Introduction to IR

- Examples:
 - Searching our email
 - Searching corporate documents
 - Searching personal medical records
 - ...

Introduction to IR

- Ranked retrieval:
 - The retriever returns the **top-k** documents **sorted** by relevance.

Introduction to IR

- Key elements:
 - Query (a clue to the information need)
 - Collection
 - Rank

Introduction to IR



- [menti.com](https://www.menti.com)

Outline

- Introduction to IR
- **IR pipeline**
- RAG
- IR Evaluation

IR pipeline

1. Vectorization
2. Indexing
3. Ranking/Retrieval

IR pipeline

- Vectorization

- Definition:

- The process of converting data (like text) into numbers

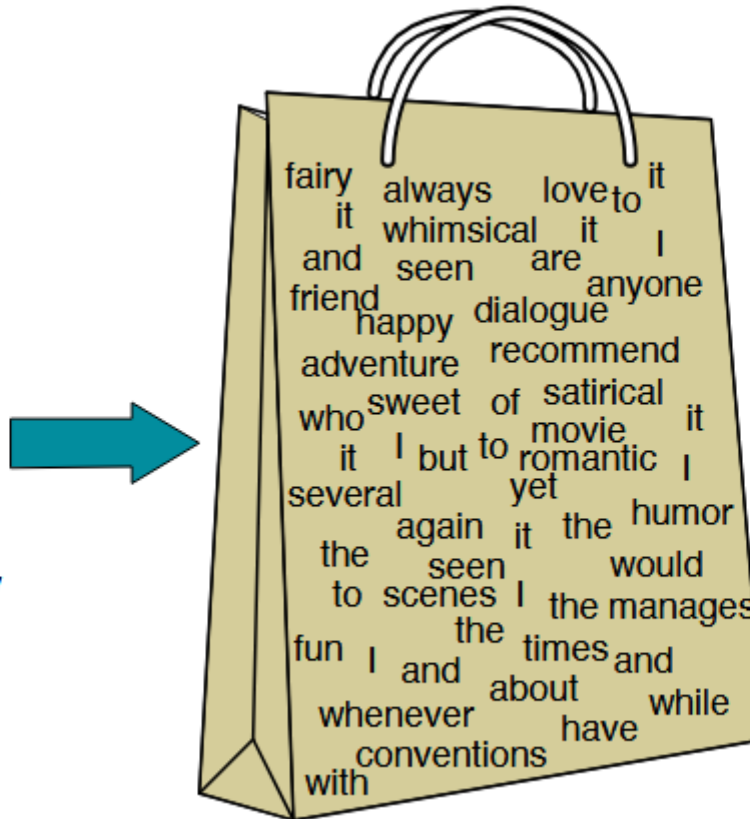
- Key aspects:

- Reduce storage space
 - Keep the important information in the document
 - Facilitate comparison

IR pipeline

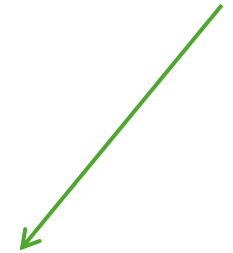
- Vectorization
 - Bag-of-words model

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



adventure	1
and	3
fairy	1
genre	1
great	1
have	1
humor	1
I	5
it	6
satirical	1
seen	2
sweet	1
the	4
times	1
to	3
whimsical	1
would	1
yet	1
...	...

Document vector
(encoded document)

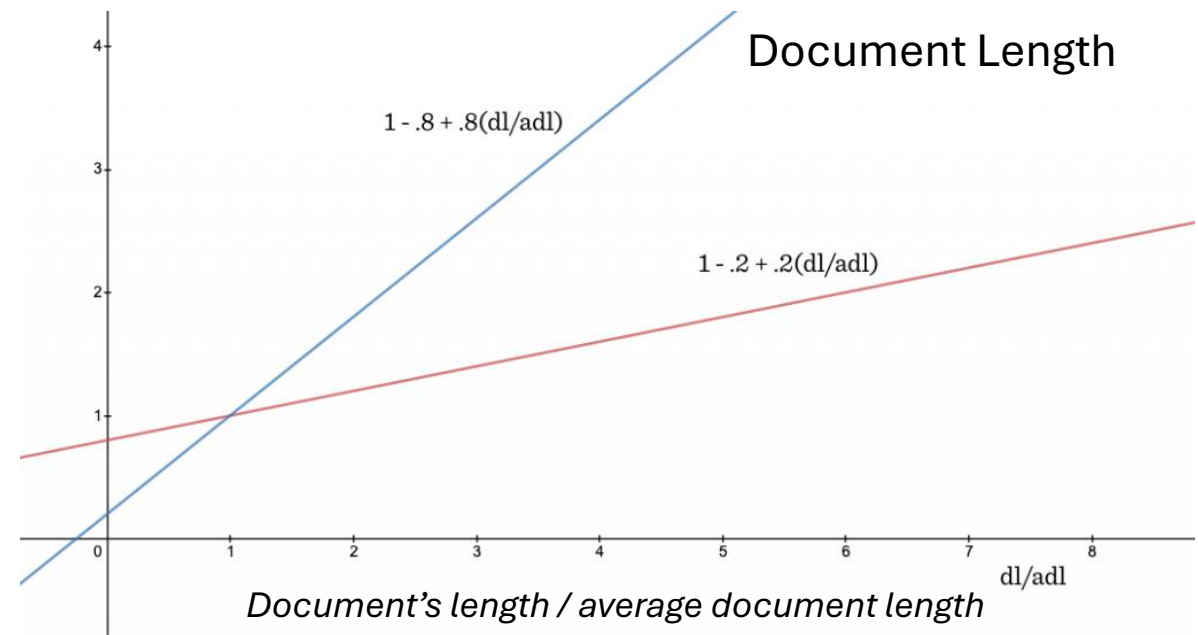
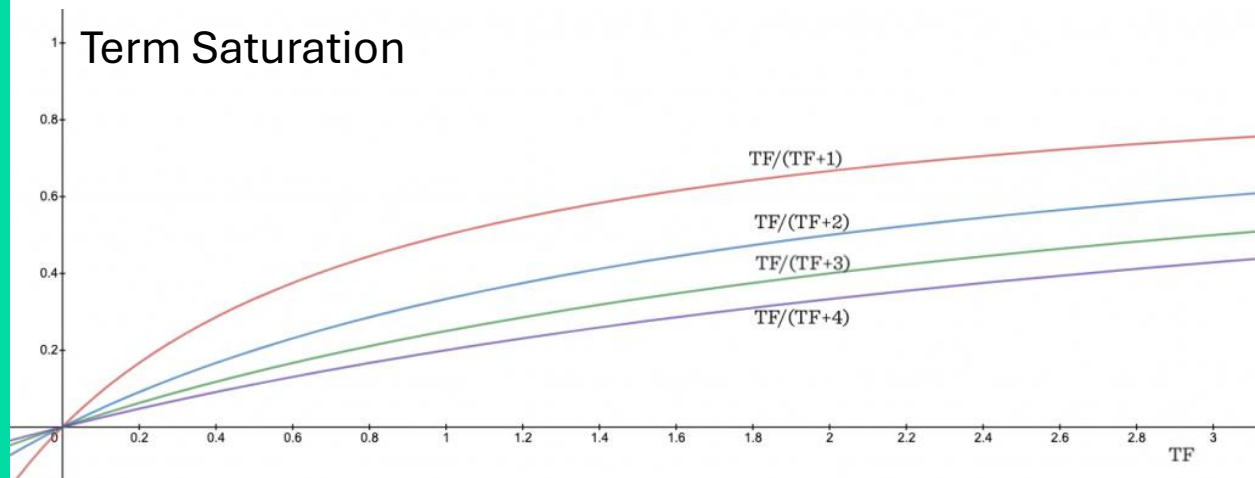


IR pipeline

- Vectorization
 - BoW
 - Binary Vector
 - Count Vector
 - TF-IDF
 - BM25
 - Word Embeddings

IR pipeline

- Vectorization
 - BM25 (*Best Matching*)
 - Motivation:
 - Long documents > term Saturation



IR pipeline

- Vectorization
 - BM25

$$BM25 = \sum_{t \in q} \log \left[\frac{N}{df(t)} \right] \cdot \frac{(k_1 + 1) \cdot tf(t, d)}{k_1 \cdot \left[(1 - b) + b \cdot \frac{dl(d)}{dl_{avg}} \right] + tf(t, d)}$$

The equation is annotated with two boxes: a green box labeled "IDF" containing the term $\log \left[\frac{N}{df(t)} \right]$, and a blue box labeled "TF saturation trick" containing the fraction $\frac{(k_1 + 1) \cdot tf(t, d)}{k_1 \cdot \left[(1 - b) + b \cdot \frac{dl(d)}{dl_{avg}} \right] + tf(t, d)}$. Within the blue box, the denominator's bracketed term is highlighted with a grey background.

Adjust saturation curve
based on document length

- k_1, b – parameters
- $dl(d)$ – length of document d
- dl_{avg} – average document length

IR pipeline

- Vectorization

Technique	Accuracy	Computation Time	Memory Usage
Bag of Words (BoW)	Low	Low	High
TF-IDF	Moderate	Moderate	High
Count Vectorizer	Low	Low	High
Word Embeddings	High	High	Moderate to High

IR pipeline

- Indexing

- Definition:

- The process of **parsing** and **organising** data into a **structured format** (an index) to enable fast, accurate search results

- Key aspects:

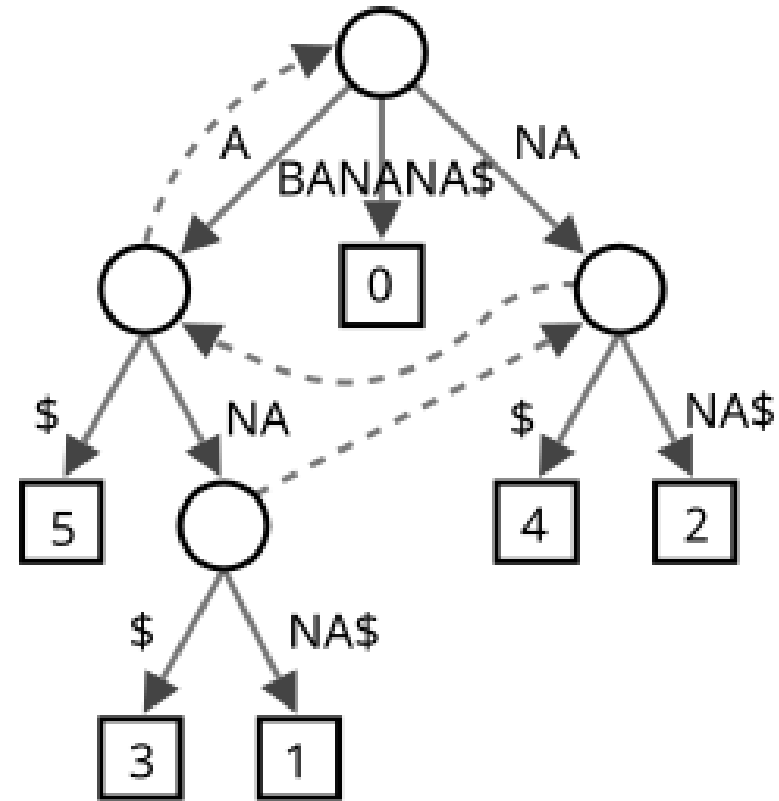
- Maximize query efficiency
 - Reduce storage space
 - Efficient access to (encoded) documents

IR pipeline

- Indexing
 - Index data structures
 - Suffix tree
 - Inverted index
 - Forward index

IR pipeline

- Indexing
 - Index data structures
 - **Suffix tree**
 - Inverted index
 - Forward index



IR pipeline

- Indexing
 - Index data structures
 - Suffix tree
 - **Inverted index**
 - Forward index

Inverted index

Word	Documents
the	Document 1, Document 3, Document 4, Document 5, Document 7
cow	Document 2, Document 3, Document 4
says	Document 5
moo	Document 7

IR pipeline

- Indexing
 - Index data structures
 - Suffix tree
 - Inverted index
 - **Forward index**

Forward Index

Document	Words
Document 1	the,cow,says,moo
Document 2	the,cat,and,the,hut
Document 3	the,dish,ran,away,with,the,spoon

IR pipeline

- Indexing
 - Example of output:

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

IR pipeline

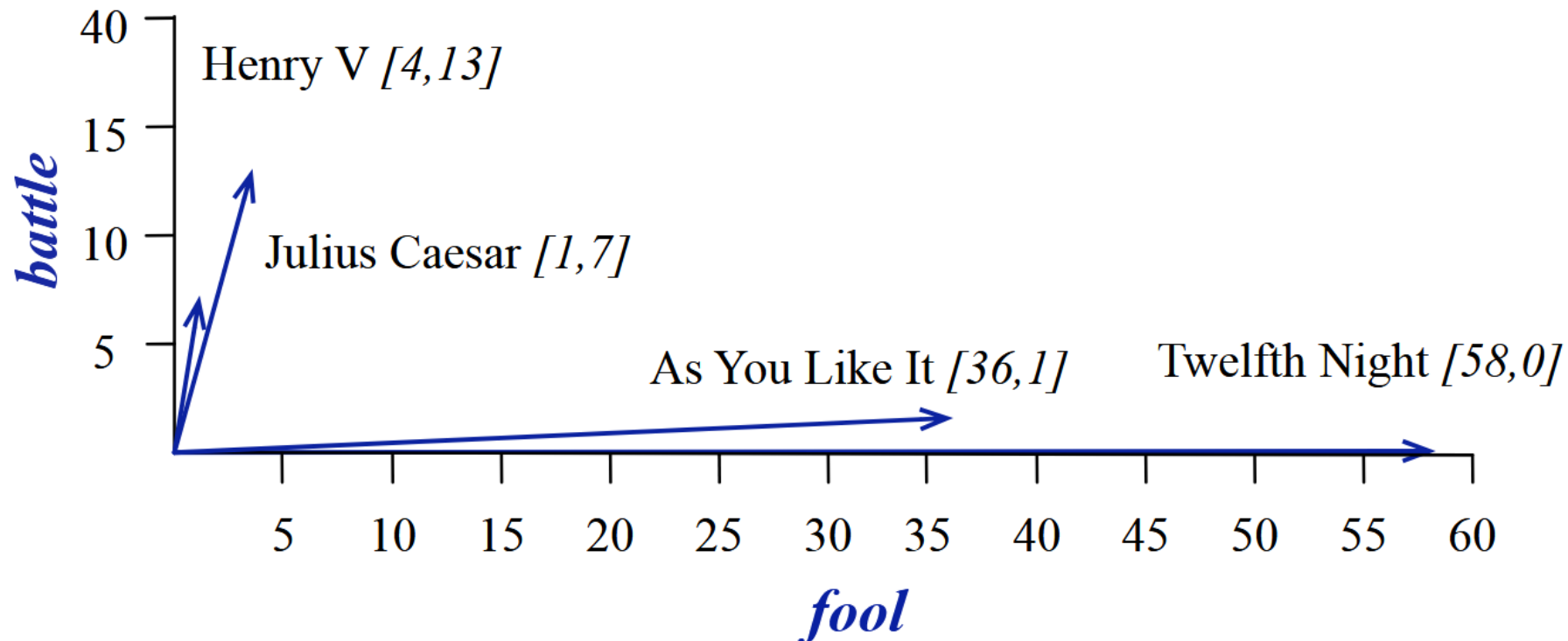
- Ranking/Retrieval
 - the process of **finding a subset of relevant items** from a large dataset, while **ranking orders** those items by relevance or importance

IR pipeline

- Ranking/Retrieval

- Example:

- The two-dimensional space [battle, fool]



IR pipeline

- Ranking/Retrieval
 - Similarity between Query and Document

	d_1	d_2	d_3	d_4	q
	As You Like It	Twelfth Night	Julius Caesar	Henry V	Query
battle	1	0	7	13	0
good	114	80	62	89	0
fool	36	58	1	4	1
wit	20	15	2	3	1

IR pipeline

- Ranking/Retrieval
 - Similarity between Query and Document

	d_1	d_2	d_3	d_4	q
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$$\text{score}(q, d) = \cos(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{|\mathbf{q}| |\mathbf{d}|}$$

IR pipeline

- Ranking/Retrieval
 - Similarity between Query and Document

$$\text{score}(q, d) = \cos(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{|\mathbf{q}| |\mathbf{d}|}$$

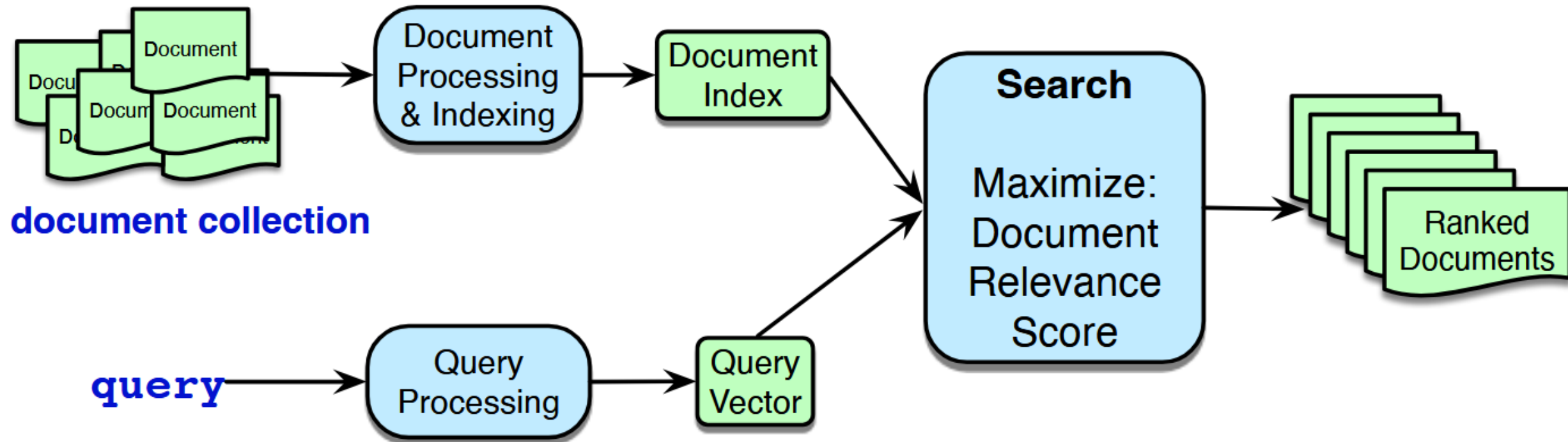
$$= \frac{\mathbf{q}}{|\mathbf{q}|} \cdot \frac{\mathbf{d}}{|\mathbf{d}|}$$

$$= \sum_{t \in \mathbf{q}} \frac{\text{tf-idf}(t, q)}{\sqrt{\sum_{q_i \in q} \text{tf-idf}^2(q_i, q)}} \cdot \frac{\text{tf-idf}(t, d)}{\sqrt{\sum_{d_i \in d} \text{tf-idf}^2(d_i, d)}}$$

IR pipeline



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IR pipeline

- Example
 - Query: sweet love
 - Collection:
 - Doc 1: Sweet sweet nurse! Love?
 - Doc 2: Sweet sorrow
 - Doc 3: How sweet is love?
 - Doc 4: Nurse!

IR pipeline

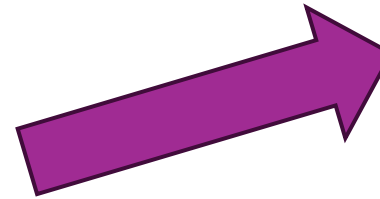
- Example
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 - Doc 3: How sweet is love?
 - Doc 4: Nurse!

IR pipeline

- Example

- Collection:

- Doc 1: Sweet sweet nurse! Love?
 - Doc 2: Sweet sorrow
 - Doc 3: How sweet is love?
 - Doc 4: Nurse!



Tokens
Sweet
Nurse
Love
Sorrow
How
Is



D1	Tf-IDF/ q
Sweet	0.357
Nurse	0.661
Love	0.661
Sorrow	0
How	0
Is	0



D2	Tf-IDF/ q
Sweet	0.203
Nurse	0
Love	0
Sorrow	0.979
How	0
Is	0

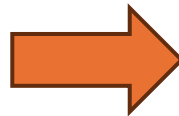
IR pipeline

- Example

- Collection:

- Doc 1: Sweet sweet nurse! Love?
 - Doc 2: Sweet sorrow
 - Doc 3: How sweet is love?
 - Doc 4: Nurse!

- Query: sweet love



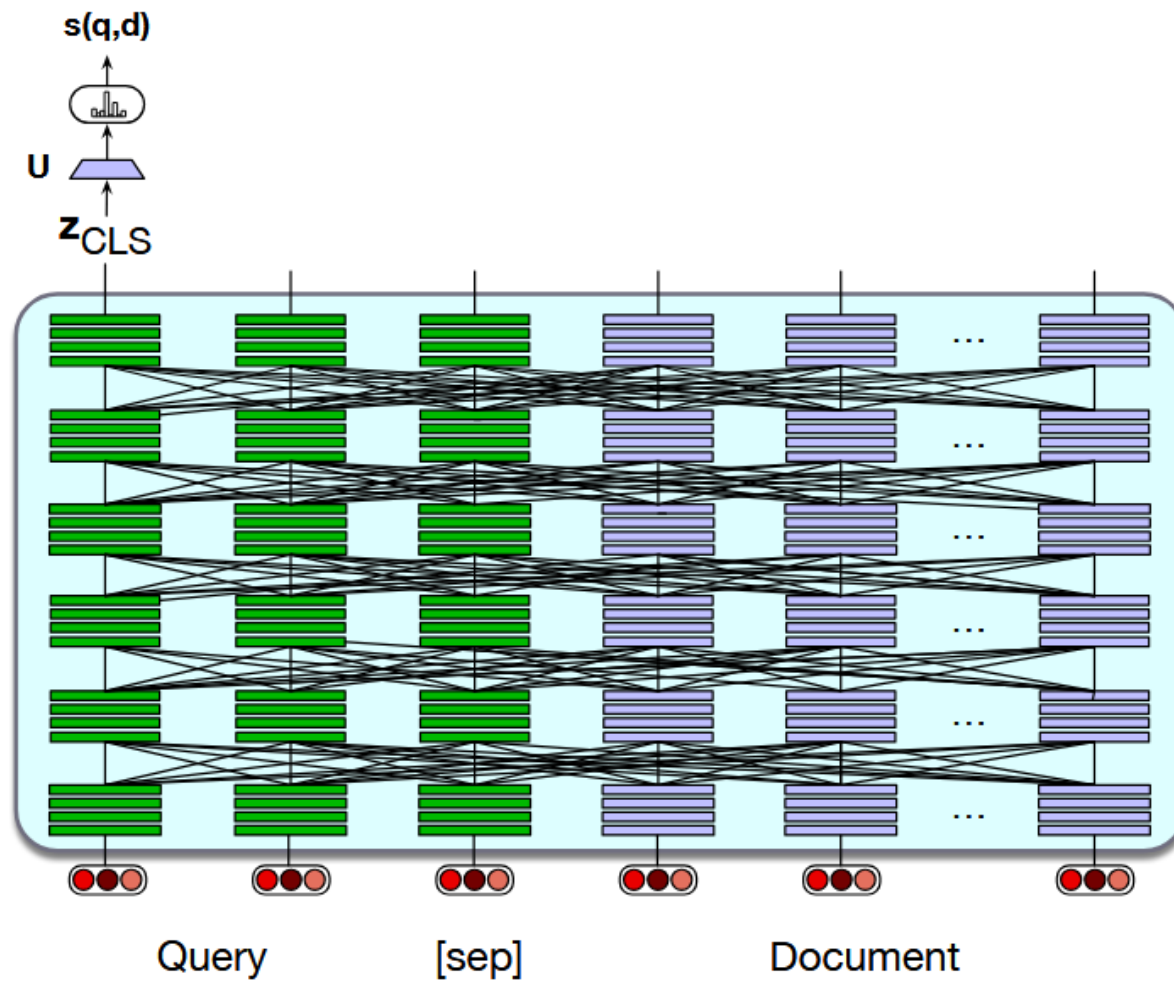
D1	Tf-IDF/ q
Sweet	0.383
Nurse	0
Love	0.924
Sorrow	0
How	0
Is	0

IR pipeline

- Example

COS $\left(\begin{array}{cc} \text{D1} & \text{Query} \\ 0.357 & 0.383 \\ 0.661 & 0 \\ 0.661 & 0.924 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{array} \right) = 0.747$ **COS** $\left(\begin{array}{cc} \text{D2} & \text{Query} \\ 0.203 & 0.383 \\ 0 & 0 \\ 0 & 0.924 \\ 0.979 & 0 \\ 0 & 0 \\ 0 & 0 \end{array} \right) = 0.078$

IR pipeline



We must rank every document for its similarity to the query!

Outline

- Introduction to IR
- IR pipeline
- **RAG**
- IR Evaluation

RAG

- Problem with classic IR
 - The vocabulary mismatch problem
- Solution:
 - Replace count vectors/TF-IDF/BM25 with **contextual embeddings**

RAG

- Retrieval-Augmented Generation

What is RAG?

Retrieving relevant data and generating accurate, context-aware responses to improve AI outputs.

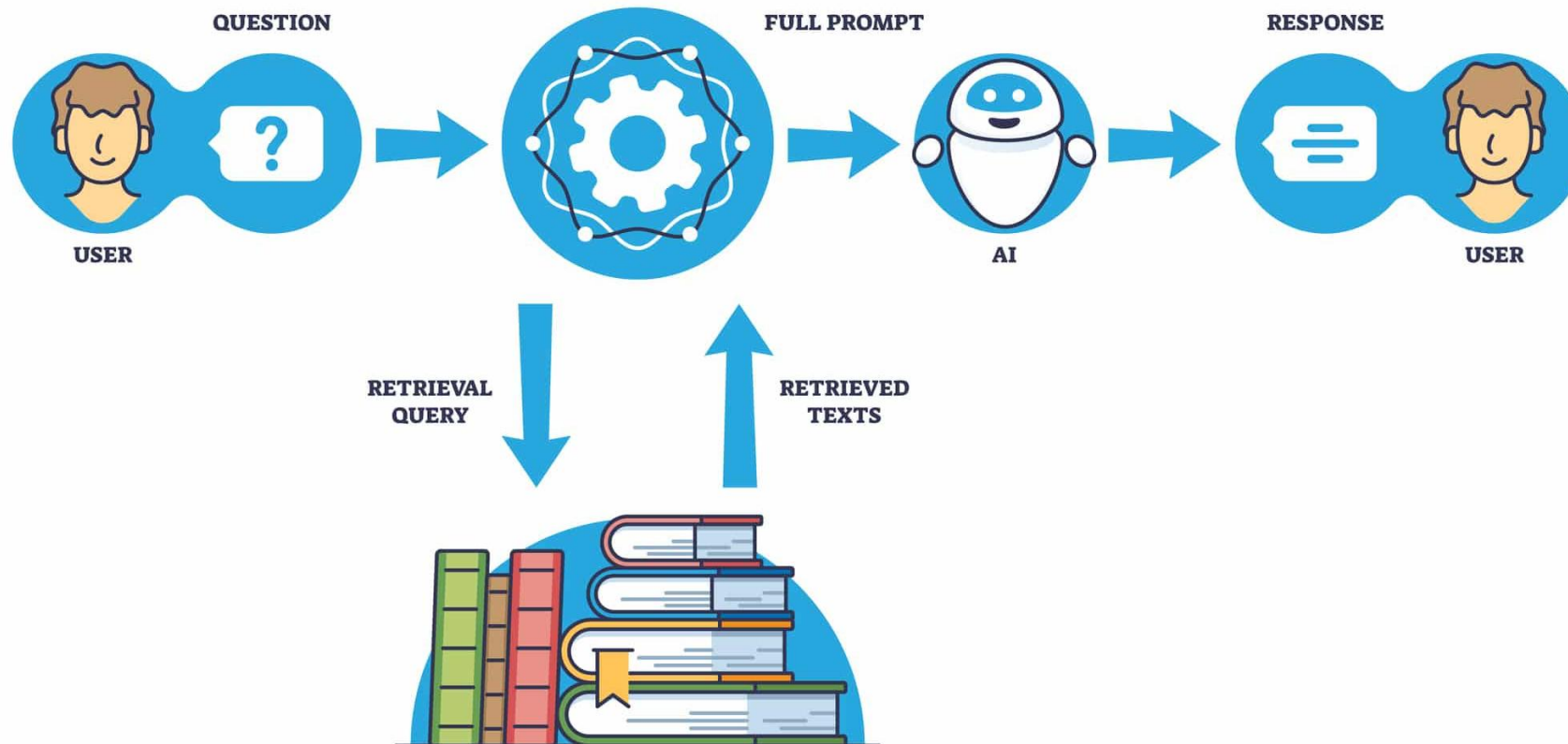
R Retrieve | Find useful information

A Augment | Add it to the AI's knowledge

G Generate | Create a better response

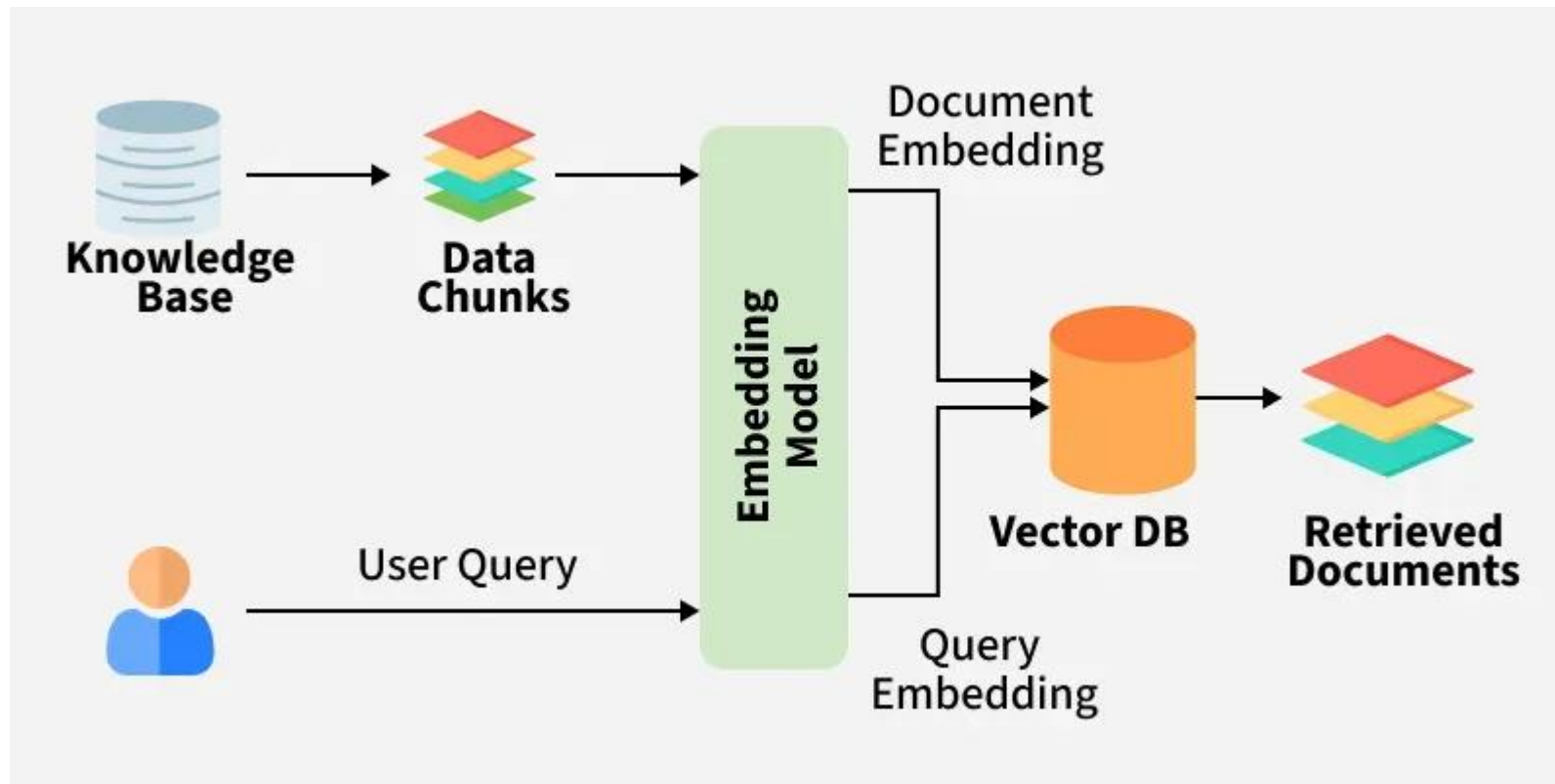
RAG

- RAG as IR pipeline



RAG

- RAG as IR pipeline



RAG

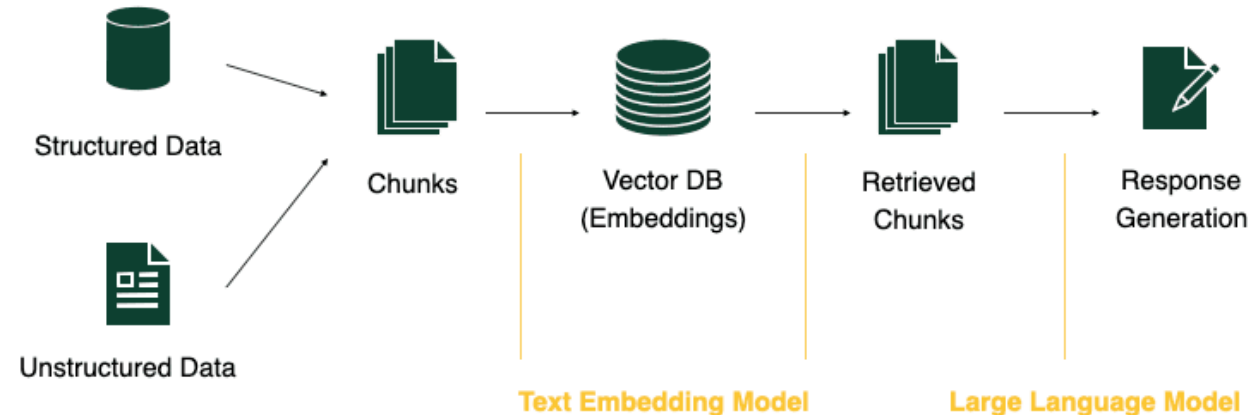
- Types of RAG
 - Simple RAG
 - Hybrid RAG
 - Graph RAG
 - Agentic RAG

RAG

- Types of RAG

- **Simple RAG**

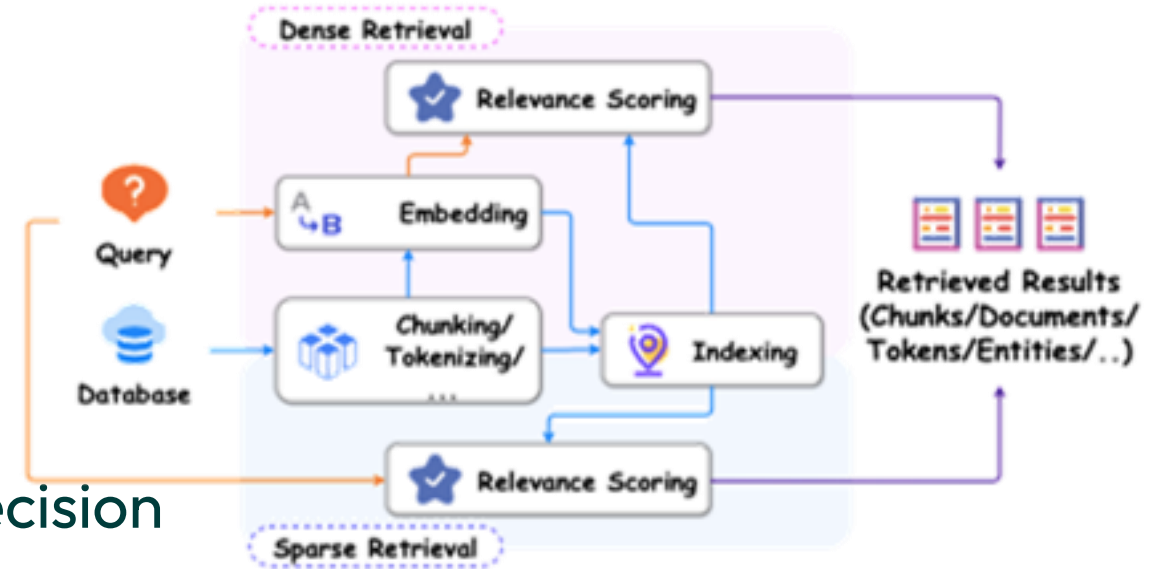
- Base architecture
 - Main advantage: simplicity



RAG

- Types of RAG
 - **Hybrid RAG**

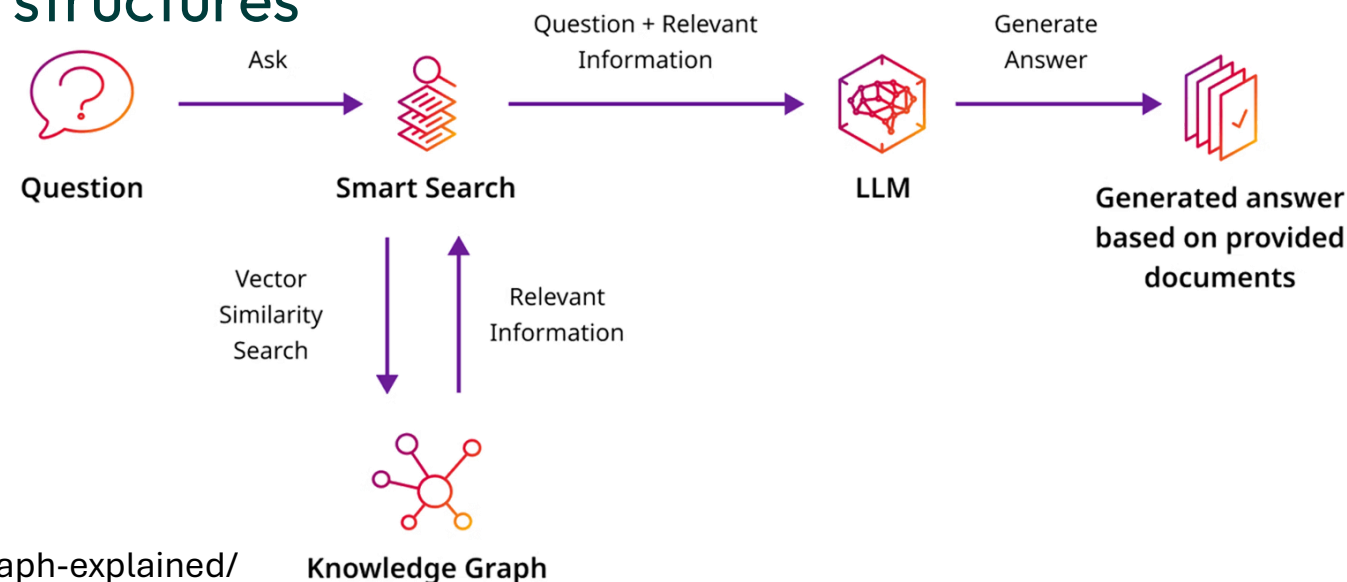
- multiple retrieval technique
 - E.g., BM25 + dense retrieval
- Advantage: improves retrieval precision



RAG

- Types of RAG
 - **Graph RAG**

- graph-based data structures to enhance retrieval
 - use of graph databases
- Advantage: structured information
- Limitation: reliance on graph structures

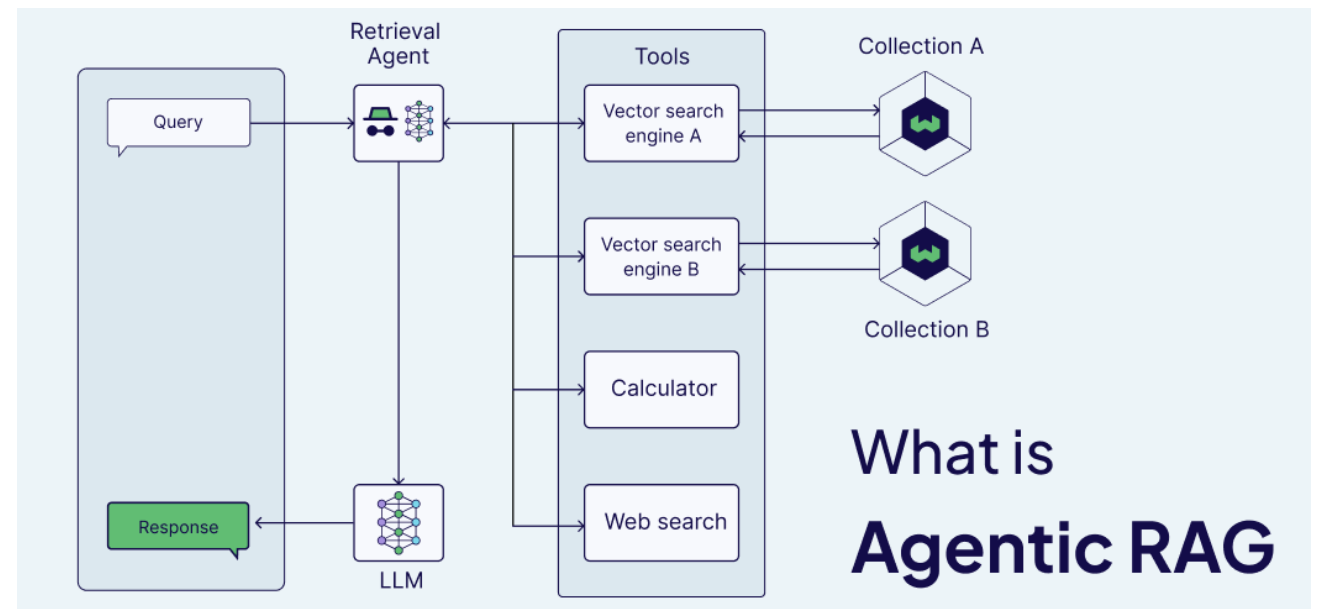


RAG

- Types of RAG

- **Agentic RAG**

- multi-agent architecture
 - Advantage: adaptability and flexibility



RAG

- Types of RAG: use cases
 - Simple RAG
 - FAQs, simple summarization
 - Hybrid RAG
 - Multi-domain queries, diverse data environments
 - Graph RAG
 - Knowledge graph queries, entity relationships, domain specialised
 - Agentic RAG
 - multiple sources, multi-step reasoning

RAG

• Types of RAG

Method	Strengths	Weaknesses
Simple RAG	Straightforward, low resource usage	Limited in handling complex queries
Hybrid RAG	Combines multiple retrieval methods	High computational cost, complex setup
AgenticRAG	Multi-agent adaptability for complex tasks	High complexity, coordination challenges
GraphRAG	Graph-based precision for structured data	Scalability issues, computational overhead

RAG

- Challenges

- Complexity:

- careful tuning and optimization

- Latency:

- challenging to deploy RAG models in real time applications.

- Quality of Retrieval:

- performance heavily depends on the quality of the retrieved documents

- Bias and Fairness:

- It can inherit biases present in the training data or retrieved documents

Outline

- Introduction to IR
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- IR Evaluation

IR Evaluation

gold standard labels

		gold positive	gold negative
<i>system output labels</i>	system positive	true positive	false positive
	system negative	false negative	true negative

IR Evaluation



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gold standard labels

		gold positive	gold negative	
<i>system output labels</i>	system positive	true positive	false positive	precision = $\frac{tp}{tp+fp}$
	system negative	false negative	true negative	
		recall = $\frac{tp}{tp+fn}$		accuracy = $\frac{tp+tn}{tp+fp+tn+fn}$

IR Evaluation

- **Precision:** % of selected items that are correct

$$\begin{aligned} & \frac{|\text{relevant retrieved docs}|}{|\text{relevant retrieved docs}| + |\text{irrelevant retrieved docs}|} \end{aligned}$$

- **Recall:** % of correct items that are selected

$$\begin{aligned} & \frac{|\text{retrieved relevant documents}|}{|\text{all relevant document}|} \end{aligned}$$

IR Evaluation

- Precision/recall are not enough
 - This is ranked retrieval



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Rank	Judgment	Precision _{Rank}	Recall _{Rank}
1	R	1.0	.11
2	N	.50	.11
3	R	.66	.22
4	N	.50	.22
5	R	.60	.33
6	R	.66	.44
7	N	.57	.44
8	R	.63	.55
9	N	.55	.55
10	N	.50	.55
11	R	.55	.66
12	N	.50	.66
13	N	.46	.66
14	N	.43	.66
15	R	.47	.77
16	N	.44	.77
17	N	.44	.77
18	R	.44	.88
19	N	.42	.88
20	N	.40	.88
21	N	.38	.88
22	N	.36	.88
23	N	.35	.88
24	N	.33	.88
25	R	.36	1.0

IR Evaluation

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10	N	.50	.55
11	R	.55	.66
12	N	.50	.66
13	N	.46	.66
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24	N	.33	.88
25	R	.36	1.0

-
- This figure shows a Precision-Recall curve for the 'recall' metric. The x-axis represents Recall (0.0 to 1.0) and the y-axis represents Precision (0.0 to 1.0). The curve starts at a precision of 1.0 for a recall of approximately 0.1. It then drops to a precision of approximately 0.5 for a recall of 0.1. The curve generally trends upwards and to the right, indicating that as recall increases, precision also tends to increase, though with some fluctuations. There are several vertical segments, indicating that multiple models achieved the same recall but with different precision levels. The curve ends at a recall of 1.0 with a precision of approximately 0.35.
- | Recall | Precision |
|--------|-----------|
| 0.1 | 1.0 |
| 0.1 | 0.5 |
| 0.22 | 0.66 |
| 0.22 | 0.5 |
| 0.33 | 0.6 |
| 0.44 | 0.66 |
| 0.44 | 0.57 |
| 0.55 | 0.63 |
| 0.55 | 0.55 |
| 0.55 | 0.5 |
| 0.67 | 0.55 |
| 0.67 | 0.5 |
| 0.67 | 0.46 |
| 0.67 | 0.43 |
| 0.67 | 0.42 |
| 0.77 | 0.47 |
| 0.77 | 0.44 |
| 0.88 | 0.44 |
| 0.88 | 0.43 |
| 0.88 | 0.42 |
| 0.88 | 0.41 |
| 0.88 | 0.40 |
| 0.88 | 0.39 |
| 0.88 | 0.38 |
| 0.88 | 0.37 |
| 0.88 | 0.36 |
| 0.88 | 0.35 |
| 1.0 | 0.35 |

Rank	Judgment	Precision _{Rank}	Recall _{Rank}
1	R	1.0	.11
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9	N	.55	.55
10	N	.50	.55
11	R	.55	.66
12	N	.50	.66
13	N	.46	.66
14	N	.43	.66
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IR Evaluation

- Precision/recall are not enough
 - Need a metric that aggregates over many queries

IR Evaluation

- Precision/recall are not enough
 - Need a metric that aggregates over many queries
 - Mean Average Precision

$$\text{MAP} = \frac{1}{|Q|} \sum_{q \in Q} \text{AP}(q) \longrightarrow \text{AP} = \frac{1}{|R_r|} \sum_{d \in R_r} \text{Precision}_r(d)$$

set of relevant documents

Conclusion

- By the end of this session, students will be able to:
 - Compare sparse and dense retrieval methods, including BM25 and embedding-based approaches, in terms of trade-offs.
 - Apply vectorisation, indexing, and similarity scoring to retrieve and rank documents.
 - Explain how RAG integrates retrieval and generation.
 - Evaluate IR architectures